L.J Institute of Engineering and Technology

Data Structures using JAVA PRACTICE BOOK (SEM-II EVEN_2025-CE RELATED BRANCHES)

Note :

This practice book is only for reference purpose . L.J.U Test question paper may not be completely set from this practice bank.

Date : 25 Apr 2025

Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
1	4	A linear list of elements in which deletion can be done from one end (front) and insertion can take place only at the other end (rear) is known as _____	A	1	Queue	Stack	Tree	Linked list
2	4	A queue follows _____	A	1	FIFO (First In First Out) principle	LIFO (Last In First Out) principle	Ordered array	Linear tree
3	4	A normal queue, if implemented using an array of size MAX_SIZE, gets full when?	A	1	Rear = MAX_SIZE – 1	Front = (rear + 1)mod MAX_SIZE	Front = rear + 1	Rear = front
4	4	What is the term for inserting into a full queue known as?	A	1	overflow	underflow	null pointer exception	program won't be compiled
5	4	Which among the following data structure may give overflow error, even though the current number of elements in it is less than its size.	D	1	Circular queue	Stack	Adaptive queue	Simple queue
6	4	Queues serve major role in ____	C	1	Simulation of recursion	Simulation of heap sort	Simulation of limited resource allocation	None of these
7	4	Write an algorithm to perform Enqueue operation in simple Queue		3				
8	4	Write an algorithm to perform Dequeue operation in simple Queue		3				
9	4	Write algorithm for INSERT, DELETE and DISPLAY function of the QUEUE.		7				
10	4	If the elements "A", "B", "C" and "D" are placed in a queue and are deleted one at a time, in what order will they be removed?	A	1	ABCD	DCBA	DCAB	ABDC
11	4	Suppose implementation supports an instruction REVERSE, which reverses the order of elements on the stack, in addition to the PUSH and POP instructions. Which one of the following statements is TRUE with respect to this modified stack?	C	1	A queue cannot be implemented using this stack.	A queue can be implemented where ENQUEUE takes a single instruction and DEQUEUE takes a sequence of two instructions.	A queue can be implemented where ENQUEUE takes a sequence of three instructions and DEQUEUE takes a single instruction.	A queue can be implemented where both ENQUEUE and DEQUEUE take a single instruction each.
12	4	Match the following:- A. Linear Queue (i). delete element from queue B. Circular Queue (ii). If (R == size – 1), Queue is full. C. Enqueue (iii). If (F== R+1 || (R==size-1 && F==0)), Queue is full D. Dequeue (iv). Insert element into queue	D	1	A-(iii), B-(ii), C-(iv), D-(i)	A-(iv), B-(iii), C-(ii), D-(i)	A-(i), B-(iii), C-(iv), D-(ii)	A-(ii), B-(iii), C-(iv), D-(i)

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13	4	Suppose you are given an implementation of a queue of integers. The operations that can be performed on the queue are:
		i. isEmpty (Q) — returns true if the queue is empty, false otherwise.
		ii. delete (Q) — deletes the element at the front of the queue and returns its value.
		iii. insert (Q, i) — inserts the integer i at the rear of the queue.
		Consider the following function:
		void f (queue Q) {
		int i;
		if(!isEmpty(Q)){
		i = delete(Q);
		f(Q); }
		insert(Q, i);
		} What operation is performed by the above function f ?
			B	1	Leaves the queue Q unchanged	Reverses the order of the elements in the queue Q	Deletes the element at the front of the queue Q and inserts it at the rear keeping the other elements in the same order	Empties the queue Q

14	4	Queue – enqueue() Array implementation. Assume that the queue is having atleat one element.
		void enqueue (int data)
		{
		if (capacity-1== rear)
		return;
		else{
		arr[rear]=data;
		return;}
		}
			B	1	The code might try to insert elements even when the queue is full which will lead to an overflow	the code overwrites existing record when we try to insert new data	the code doesn't check if the data is empty	the code doesn't update the front of the queue

15	4	Provided the space is available, then to insert an element x in the queue, we can use for the following class
		class queue
		{
		int Q[] = new int[20];
		int f=-1, r=-1;
		}
		Then which of the options is correct?
			B	1	++f ; ++Q[r]=x	++f ; Q[++r]=x	++f ; Q[r++]=x	None of the above

16	4	Consider a standard Circular Queue 'q' implementation (which has the same condition for Queue Full and Queue Empty) whose size is 11 and the elements of the queue are q[0], q[1], q[2].....,q[10]. The front and rear pointers are initialized to point at q[2] . In which position will the 8th element be added?	D	1	q[0]	q[1]	q[2]	q[10]
17	4	Write a JAVA functions for insertion and deletion operation in simple queue.		7				

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18	4	Following is pseudo code of a function that takes a Queue as an argument, and uses a stack S to do processing.
		void fun(Queue Q)
		{
		Stack S; // Say it creates an empty stack S
		// Run while Q is not empty
		while (!isEmpty(Q))
		{
		// deQueue an item from Q and push the dequeued item to S
		push(S, deQueue(Q));
		}
		// Run while Stack S is not empty
		while (!isEmpty(S))
		{
		// Pop an item from S and enqueue the poppped item to Q
		enQueue(Q, pop(S));
		}
		}
		What does the above function do in general?
			D	1	Removes the last from Q	Keeps the Q same as it was before the call	Makes Q empty	Reverses the Q

19	4	Consider the following pseudo code. Assume that IntQueue is an integer queue. What does the function fun do?
		void fun(int n)
		{
		IntQueue Q // creates new integer queue
		enqueue(Q,0);
		enqueue(Q,1);
		for (int i = 0; i < n; i++)
		{
		int a = dequeue(Q);
		int b = dequeue(Q);
		enqueue(Q, b);
		enqueue(Q, a + b);
		print(a);
		}
		}
			C	1	Prints numbers from 0 to n-1	Prints numbers from n-1 to 0	Prints first n Fibonacci numbers	Prints first n Fibonacci numbers in reverse order

20	4	Consider the following pseudo code implementation of Enqueue and Dequeue operations using two initially empty very large stacks P and Q, with primitive stack operations PUSH and POP.
		Stack P, Q;
		Enqueue(key k)
		PUSH(k,P);
		Dequeue( ) {
		if(Q is empty)
		{
		while(P is not empty)
		MISSING STATEMENT ;
		}
		return POP(Q);
		}
		Choose the correct statement in place of MISSING STATEMENT. (Ignore error handling).
			A	1	PUSH(POP(P),Q)	PUSH(POP(P))	PUSH(P)	PUSH(POP(Q))

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21	4	Let Q be a Queue. enqueue and dequeue are usual Queue operations to add and delete element respectively.
		Q.enqueue(x) adds an element x to the queue.
		Q.dequeue() performs a dequeue operation on the queue and returns the value that gets dequeued.
		Consider the following segment of code:
		
		Q.enqueue(1);
		int count=1;
		do {
		count=count + 1;
		x = Q. dequeue();
		Q.enqueue (2*x);
		Q.enqueue(4*x);
		} while (x!= 32);
		What will be the value of the variable count, when the above segment of code completes its execution?
			C	1	11	10	12	13

22	4	Write a program to perform insert and delete routines on a queue.		7				
23	4	Suppose a queue is maintained by a circular array holding 10 elements. Find the number of elements in the queue after the following operations 1) front=3, rear=7 2) front=9, rear=4 3) Front =4, rear=5 and then two elements are deleted.	A	1	0	1	2	3
24	4	Consider the following statements: i. First-in-first out types of computations are efficiently supported by STACKS. ii. Implementing QUEUES on a circular array is more efficient than implementing QUEUES on a linear array with two indices. iii. Last-in-first-out type of computations are efficiently supported by QUEUES. Which of the following is correct?	A	1	Only (ii) is true	(i) and (ii) are true	(iii) and (iv) are true	Only (iii) is true
25	4	In a circular queue, how do you increment the rear end of the queue?	B	1	rear++	(rear+1) % CAPACITY	(rear % CAPACITY)+1	rear–
26	4	What is the need for a circular queue?	A	1	effective usage of memory	easier computations	to delete elements based on priority	implement LIFO principle in queues
27	4	Let the following circular queue can accommodate maximum six elements with the following data front = 2 rear = 4 queue = __, L, M, N, ___, ___ What will happen after ADD O operation takes place?	A	1	front = 2 rear = 5 queue = _____; L, M, N, O, ___	front = 3 rear = 5 queue = L, M, N, O, ___	front = 3 rear = 4 queue = _____; L, M, N, O, ___	front = 2 rear = 4 queue = L, M, N, O, ___
28	4	A circular queue is implemented using an array of size 10. The array index starts with 0, front is 6, and rear is 9. The insertion of next element takes place at the array index.	A	1	0	7	9	10
29	4	Given a circular queue implemented with an array of size 8 (indexing from 0 to 7) and F=2 & R=7. After inserting an element, what would the new position of the rear pointer?	B	0.5	1	0	8	3

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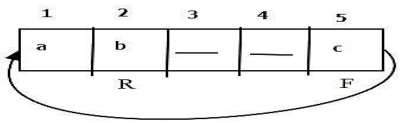
30	4	Suppose a queue is implemented using a circular array QUEUE with N=20 memory cells. The queue is initially empty. Perform the following sequence of operations: Enqueue 7 elements Dequeue 5 elements Enqueue 12 elements Dequeue 10 elements Enqueue 4 elements Dequeue 8 elements Enqueue 3 elements Dequeue 2 elements What is the number of elements in QUEUE at the end?	B	1	0	1	2	3
31	4	Suppose we have a standard circular queue named q of size 36 with Two different initial pointer positions Case 1 front = 5, rear = 9 and Case 2 front = 24, rear = 15. Now, we perform insertion 12 times then where would the 12th element go into this queue after an enqueue operation? consider indexing starts from 0.	A	1	Case 1 – q[21], Case 2 – OVERFLOW	Case 1 – q[20], Case 2 – q[27]	Case 1 – q[21], Case 2 – q[26]	Case 1 – q[20], Case 2 – q[26]
32	4	Write an algorithm to perform Enqueue operation in Circular Queue		4				
33	4	Write an algorithm to perform Dequeue operation in Circular Queue		4				
34	4	Circular Queue is also known as _____	A	1	Ring Buffer	Square Buffer	Rectangle Buffer	Curve Buffer
35	4	If the MAX_SIZE is the size of the array used in the implementation of circular queue. How is rear manipulated while inserting an element in the queue?	C	1	rear=(rear%1)+MAX_SIZE	rear=rear%(MAX_SIZE+1)	rear=(rear+1)%MAX_SIZE	rear=rear+(1%MAX_SIZE)
36	4	If the MAX_SIZE is the size of the array used in the implementation of circular queue, array index start with 0, front point to the first element in the queue, and rear point to the last element in the queue. Which of the following condition specify that circular queue is FULL?	B	1	Front=rear= -1	Front=(rear+1)%MAX_SIZE	Rear=front+1	Rear=(front+1)%MAX_SIZE
37	4	If the MAX_SIZE is the size of the array used in the implementation of circular queue, array index start with 0, front point to the first element in the queue, and rear point to the last element in the queue. Which of the following condition specify that circular queue is EMPTY?	B	1	Front=rear=0	Front= rear=-1	Front=rear+1	Front=(rear+1)%MAX_SIZE
38	4	An array of size MAX_SIZE is used to implement a circular queue. Front, Rear, and count are tracked. Suppose front is 0 and rear is MAX_SIZE -1. How many elements are present in the queue?	D	1	Zero	One	MAX_SIZE-1	MAX_SIZE
39	4	The initial configuration of circular queue as follows  What is status of states of queue contents after the following sequence of steps enqueue x, dequeue, enqueue y, dequeue, dequeue	C	1	x,y,____,_____,_____	x,___,y,____,____	____,_____, x ,y,____	_____,x,y,_____,_____

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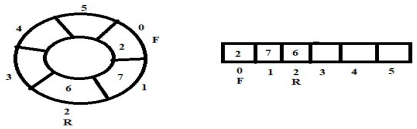
40	4	Consider a circular queue of size 6. Let Front =2, Rear =4, and Queue : __, L, M, N, __, __ Describe the queue as following operations are performed. 1) Add O 2) Add P 3) Delete 4) Delete 5) Add Q, R, S 6) Delete	B	1	R=4 and F=2	R=3 and F=5	R=3 and F=4	Queue Overflow
41	4	Suppose we have a circular array implementation of the queue type, with ten items in the queue stored at data [2] through data [11]. The current SIZE is 22. Where does the insert method place the new entry in the array?	C	1	data[1]	data[22]	data[12]	data[11]
42	4	The problem of false alarm about overflow of linear queue is removed by the application of_____.	D	1	deque	Stack	Array	Circular Queue
43	4	Write an algorithm for circular queue that insert an element at rear end		4				
44	4	Write a JAVA program to implement a circular queue using array with all necessary overflow and underflow checks		7				
45	4	Write a JAVA function for circular queue for the below operations. 1)To check Overflow condition 2)To check Underflow condition		2				
46	4	Write a JAVA function to implement Circular queue insertion.		2				
47	4	Consider a standard Circular Queue 'q' implementation (which has the same condition for Queue Full and Queue Empty) whose size is 11 and the elements of the queue are q[0], q[1], q[2].....,q[10]. The front and rear pointers are initialized to point at q[2] . In which position will the ninth element be added?	A	1	q[0]	q[1]	q[9]	q[10]
48	4	What will the final value of Front and Rear pointer value after given below operation for linear and circular queue.  	A	1	linear: F=3, R=5 circular: F=3, R=1	linear: F=3, R=5 circular: F=3, R=2	linear: F=5, R=5 circular: F=3, R=1	linear: F=3, R=3 circular: F=3, R=1

49	4	A circular array-based queue q is capable of holding 7 elements. After execution of the following code, find the element at index 2, if the array is initially empty and array has indices from 0 to 6.
		for (x=1; x<=6; x++)
		q.enqueue (x);
		for (x=1; x<=4; x++)
		{
		q.dequeue();
		q.enqueue (q. Dequeue ());
		}
			D	1	1	2	3	4

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50	4	Consider the following queue, where queue is a circular queue having 6 memory cells. Front=2, Rear=4 Queue: _, A, C, D, _, _ Describe queue as following operation take place: F is added to the queue Two letters are deleted R is added to the queue S is added to the queue One letter is deleted		3				
51	4	Perform following operations in a circular queue of length 4 and give the Front, Rear and Size of the queue after each operation. 1) Insert A, B 2) Insert C 3) Delete 4) Insert D		3				
52	4	Write an algorithm/program to implement Insert & Delete operation into a Circular Queue using array representation of Queue.		7				
53	5	What is a deque?	A	1	A queue with insert/delete defined for both front and rear ends of the queue	A queue implemented with a doubly linked list	A queue implemented with both singly and doubly linked lists	A queue with insert/delete defined for front side of the queue
54	5	Which of the following data structure follows FIFO technique?	D	1	Priority Queue	deque	Both of these	None of these
55	5	Consider a Double Ended Queue given below, initially empty: _ _ _ _ _ _ _ _ The following operations are performed on the dequeue: Insert X at the front. Insert Y at the rear. Insert Z at the rear. Delete an element from the front. Insert W at the front. Delete an element from the rear. What is the resulting state of the double-ended queue after performing these operations?	D	1	_ _ _ _ _ _ _ _	_ W _ X Y Z _ _	_ _ W X Y _ _ _	None
56	5	Given an initially empty Deque named D, perform the operations below in the order shown, then answer the questions below D.insertAtREAR(12) ; D.insertAtREAR(22) ; D.insertAtFront(32); D.insertAtREAR (42); D.removeFromFront(); D.insertAtREAR (52); D.removeFromFront(); D.insertAtREAR (62); D.insertAtFront (72); D.insertAtFront (82); D.removeFromREAR (); D.removeFromREAR (); (i) How many elements are in the deque? (ii) What value is at the front of the deque? (iii) What value is at the rear of the deque?	C	1	elements = 0, front = 72, rear = 52	elements = 0, front = 82, rear = 52	elements = 4, front = 82, rear = 42	None

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57	5	For a double ended queue the following method performs what?
		void display()
		{
		int i=front;
		while(i!=rear)
		{
		System.out.print(a[i]);
		i=(i+1)%size;
		} }
			B	0.5	The method displays all the elements of queue.	The method displays all the elements of queue except last.	The method displays all the elements of queue except first.	The method displays all the elements of queue in reverse

58	5	A queue where deletion can be made from both ends, but insertion can be made at one end is called .	B	0.5	output restricted queue	input restricted queue	circular queue	None of these
59	5	What is the result of applying the dequeue operation on an empty queue?	B	0.5	The queue have one element	An error is raised indicating an underflow condition.	The front element is removed, and the queue becomes empty.	None of these
60	5	Consider a deque given below which has LEFT=1, RIGHT=5 _ A B C D E _ _ _ _ . Now perform the following operations on the deque and trace values of LEFT and RIGHT respectively. 1. Add F on the left. 2. Add G on the right. 3. Add H on the right. 4. Delete two alphabets from left 5. Add I on the right	A	1	2,8	1,8	1,7	2,7
61	5	After performing these set of operations in double ended queue, what does the final list look contain? InsertFront(10); InsertFront(20); InsertRear(30); DeleteFront(); InsertRear(40); InsertRear(10); DeleteRear(); InsertRear(15); display();	D	1	10 30 10 15	20 30 40 15	20 30 40 10	10 30 40 15
62	5	A Double-ended queue supports operations such as adding and removing items from both the sides of the queue. They support four operations like addFront(adding item to top of the queue), addRear(adding item to the bottom of the queue), removeFront(removing item from the top of the queue) and removeRear(removing item from the bottom of the queue). You are given only stacks to implement this data structure. You can implement only push and pop operations. What are the total number of stacks required for this operation?(you can reuse the stack)	B	1	1	2	3	4
63	5	Consider a Double Ended Queue given below which has F=3, R=4 and size = 8 _ _ B C _ _ _ _ Now perform the following operations on the dequeue (Insert A at front, Insert D at front, Insert E at rear , Delete two alphabets from front) What is the value of F and R?	A	1	3,5	3,6	1,5	2,5

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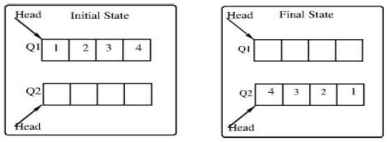
64	5	Given an initially empty Deque named D, perform the operations below in the order shown, then answer the questions below Deque < int > D; D. insertAtREAR(42) ; D. insertAtREAR(17) ; D. insertAtFront(12); D. insertAtREAR (87) ; D. removeFromFront(); D. insertAtREAR (23) ; D. removeFromFront(); D. insertAtREAR (99) D. insertAtFront (88) ; D. insertAtFront (44) ; D. removeFromREAR (); (a) How many elements are in the deque? (b) What value is at the front of the deque? (c) What value is at the rear of the deque?	D	1	element=0,f=44,r=17	element=1,f=23,r=88	element=5,f=99,r=88	element=5,f=44,r=23
65	5	Consider the double ended queues Q1 containing four elements and Q2 containing none (shown as the Initial State in the figure). The only operations allowed on these two queues are Enqueue(Q, element) and Dequeue(Q).The minimum number of Enqueue operations on Q1 required to place the elements of Q1 in Q2 in reverse order (shown as the Final State in the figure) without using any additional storage is_____.  	A	1	0	1	2	3
66	5	Write a program to implement Deque Operations		7				
67	5	Write a function to insert and delete items from double ended queue from front		4				
68	5	Write a function to insert and delete items from double ended queue from rear		4				
69	5	Write all necessary methods to develop input restricted Queue.(Do not use Link list for implementation, display method not required)		4				
70	5	If priority queue is implemeneted using arrays then which of the following is not the basic operation of the queue?	A	1	enqueue	dequeue	peek	Reverse
71	5	Which of the following can not be used to implemenet priority queue?	D	1	Arrays	Linked List	Heaps	None of these
72	5	Which one of the following is an application of Queue Data Structure?	D	1	When a resource is shared among multiple consumers.	When data is transferred asynchronously (data not necessarily received at same rate as sent) between two processes	Load Balancing	All of the above
73	6	Which of the following is not a disadvantage to the usage of array?	D	1	Fixed size	There are chances of wastage of memory space if elements inserted in an array are lesser than the allocated size	Insertion based on position	Accessing elements at specified positions
74	6	A linear collection of data elements where the linear node is given by means of reference is called?	A	1	Linked list	Node list	Primitive list	Unordered list
75	6	In linked list each node contains a minimum of two fields. One field is data field to store the data second field is?	C	1	reference to character	reference to integer	reference to node	Node

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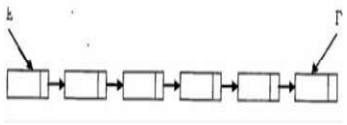
76	6	Linked list is considered as an example of _____ type of memory allocation.	A	1	Dynamic	Static	Compile time	Heap
77	6	In Linked List implementation, a node carries information regarding _____	C	1	Data	Link	Data and Link	Node
78	6	Linked list data structure offers considerable saving in _____	C	1	Computational Time	Space Utilization	Space Utilization and Computational Time	Speed Utilization
79	6	Which of the following points is/are not true about Linked List data structure when it is compared with an array?	D	1	Arrays have better cache locality that can make them better in terms of performance	It is easy to insert and delete elements in Linked List	Random access is not allowed in a typical implementation of Linked Lists	Access of elements in linked list takes less time than compared to arrays

80	6	What is the output of following function for start pointing to first node of following linked list?
		1->2->3->4->5->6
		void fun(Node start)
		{
		if(start == null)
		return;
		System.out.print(" " + start.data);
		if(start.next != null)
		fun(start.next.next);
		System.out.print(" "+ start.data);
		}
		Note - Cosnider inner class Node has outer class Linkedlist and having two data memeber data(to store the data) and next (to store the reference of the next Node). First of a given likedlist is passed as an argument to the given function
			D	1	1 4 6 6 4 1	1 3 5 1 3 5	1 2 3 5	1 3 5 5 3 1

81	6	What is the output of following function where head pointing to first node of following linked list?
		10→20→30→40→50
		void fun(Node head)
		{
		int x=45;
		System.out.print(" " +head.info);
		head=head.link.link;
		if(head.info<=x)
		System.out.print(" " +head.info);
		}
			C	1	10	10 20	10 30	None

82	6	What is the output of following function where head pointing to first node of following linked list?
		10->20->30->40->50
		void fun(Node head)
		{
		int x=25;
		System.out.print(" " +head.info);
		head=head.link.link;
		if(head.info<=x)
		System.out.print(" " +head.info);
		}
			B	1	25	10	20	20 30

83	6	Given reference to a node X in a singly linked list. Only one reference is given, reference to first node is not given, can we delete the node X from given linked list?	A	1	Possible if X is not last node	Possible if size of linked list is even	Possible if size of linked list is odd	Possible if X is not first node
84	6	A variant of linked list in which last node of the list points to null?	A	1	Singly linked list	Doubly linked list	Circular linked list	Multiply linked list
85	6	In Singly linked list each node contain _____ fields.	B	1	One	Two	Three	Four
86	6	Which of the following points is/are true about Linked List data structure when it is compared with array	D	1	Arrays have better cache locality that can make them better in terms of performance.	The size of array has to be pre-decided, linked lists can change their size any time.	Random access is not allowed in a typical implementation of Linked Lists	All of the Given

Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
87	6	You are given pointers to first and last nodes of a singly linked list, which of the following operations are dependent on the length of the linked list?	C	1	Delete the first element	Insert a new element as a first element	Delete the last element of the list	Add a new element at the end of the list
88	6	Consider a single linked list where F and L are pointers to the first and last elements respectively of the linked list. The time for performing which of the given operations depends on the length of the linked list? 	C	1	Delete the first element of the list	Interchange the first two elements of the list	Delete the last element of the list	Add an element at the end of the list

89	6	What will be the output of the following code segment if list is: 10->20->30->40->50->60?
		void solve(Node root) {
		int s = 0;
		while(root.next != null) {
		s += root.val;
		root = root.next;
		}
		System.out.println(s);}
		Cosnider the Node inner class of SLinkedList is having two data member - val, next
			C	1	120	210	150	60

90	6	Consider the Java code fragment given below.
		void join(Node m, Node n){
		Node p = n;
		while(p.next != null)
		{
		p = p.next;
		}
		p.next = m;}}
		Assuming that m and n point to valid null terminated, non empty linked lists, invocation of join will
			A	1	append list m to the end of list n	append list n to the end of list m	always gives null pointer dereference for all inputs	depends on the input

91	6	Consider the following linked list : class Node { int info; Node link; }head = null; What will be the value of the following statement? Head.link.link.link.info; 10-> 12-> 14-> 16-> 20	D	1	10	12	14	16

Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

92	6	The following Java function takes a singly-linked list as input argument. It modifies the list by moving the last element to the front of the list and returns the modified list. Some part of the code is left blank.
		Node move_to_front(Node head)
		{
		Node p, q;
		if ((head == null) || (head.next == null))
		return head;
		q = null;
		p = head;
		while (p.next != null)
		{
		q=p;
		p=p.next;
		}
		_____
		return head; }
		Choose the correct alternative to replace the blank line.
			D	1	q=null;p.next=head;head=p;	q.next=null;head=p;p.next=head;	head=p;p.next=q;q.next=null;	q.next=null;p.next=head;head=p;

93	6	What does the following code snippet do?
		int solve (ListNode list) {
		ListNode fast = list;
		ListNode slow = list;
		while(fast.next != null && fast.next.next != null) {
		fast = fast.next.next;
		slow = slow.next;
		}
		return slow.data;
		}
		}
			A	1	Find middle element in the linked list.	Find last element in the linked list	Find first element in the linked list	No elements

94	6	In the linked list implementation of the queue, where does the insert method place the new entry on the linked list?	B	1	At the head	At the tail	After all other entries that are greater than the new entry	After all other entries that are smaller than the new entry
95	6	Suppose cursor refers to a node in a linked list. What Boolean expression will be true when cursor refers to the tail node of the list?	A	1	cursor.next == null	cursor == null	cursor.data == null	cursor.data == 0

96	6	What is the output of the following function when it is called?
		void disp(node first)
		if(first != null)
		{
		System.out.println(" "+first.data);
		disp(first.next);
		}
			B	1	Display the data of next node always in the list	Display the data of all nodes of the list	Does not execute	Display the data of the first node only

Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

97	6	node start=head;
		Suppose start is a reference to the head of one SLL.
		class node{
		int data;
		node next;
		}
		
		Consider the above code and predict what will be the output by following piece of code
		System.out.println(start.next.data);
			B	1	Print the “data” field of 1st node	Print the “data” field of 2nd node	Print the “data” field of 3rd node	Print the “data” field of 4th node

98	6	What does the following function do for a given Linked List with first node as head?
		
		void fun1(Slinkedlist head)
		{
		if(head==null)
		return;
		fun1(head.next);
		System.out.println(" "+head.data);
		}
			B	1	Prints all nodes of linked lists	Prints all nodes of linked list in reverse order	Prints alternate nodes of Linked List	Prints alternate nodes in reverse order

99	6	What is the output, if a SLL:1→2→3→4→5 is passed in the above Java code if the inner class node contains int data memeber and node next member?
		void print(node ptr)
		{
		  if(ptr!= null)
		  {
		  System.out.print(ptr.data);
		  do{
		  System.out.print(ptr.data);
		  } while(ptr.next != null);
		  }
		}
		Assume Head reference at node 1
			C	1	1 2 3 4 5	5 4 3 2 1	1 1 1 1 1…..	1 2 3 4

100	6	If the Queue is implemented using singly linked list, keeping track of a front and rear reference, which of these references will change during an insertion into a non-empty queue?	D	1	Neither of the pointer change	Only front Pointer changes	Both of the pointers changes	Only rear Pointer changes

101	6	What is the output of following function for start refered to first node of following linked list? 1->2->3->4->5->6
		void fun(Node start)
		{
		  if(start == null)
		  return;
		  System.out.print(start.next.data);
		
		  if(start.next != null)
		  fun(start.next.next);
		  System.out.print(start.data);
		}
			A	1	2 4 6 5 3 1	1 3 5 5 3 1	2 4 6 6 4 2	2 4 6 3 1

Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

102	6	The following java function takes a single-linked list of integers as a parameter and rearranges the elements of the list. The function is called with the list containing the integers 1, 2, 3, 4, 5, 6, 7 in the given order. What will be the contents of the list after the function completes execution?
		class Node
		{
		int value;
		Node next;
		}
		void rearrange(Node list)
		{
		Node p,q;
		int temp;
		if ((list==null) || list.next==null)
		return;
		p = list;
		q = list.next;
		while(q != null)
		{
		temp = p.value;
		p.value = q.value;
		q.value = temp;
		p = q.next;
		if(p.next != null){
		q = p.next;}
		else { q = null;}
		}
		}
			B	1	1,2,3,4,5,6,7	2,1,4,3,6,5,7	1,3,2,5,4,7,6	2,3,4,5,6,7,1

103	6	Consider the following function to traverse a linked list. Choose the correct missing statement at the place of ? to write the function correctly for traverse the list.
		void traverse(Node head)
		{
		??????????????????
		while (t != null)
		{
		System.out.print(" "+ t.data);
		t = t.next;
		}
		}
			D	1	t=thead;	head=t;	Node=head;	Node t=head;

104	6	In linked list implementation of a stack, where does a new element be inserted?	A	1	At the head of link list	At the center position in the link list	At the tail of the link list	At any position in the linked list
105	6	In linked list implementation of a simple queue, front and rear reference of nodes are tracked. Which of these references will change during an insertion into EMPTY and NONEMPTY queue respectively?	C	1	EMPTY: Only front pointer NONEMPTY: Only front pointer	EMPTY: Only front pointer NONEMPTY: Only rear pointer	EMPTY: Both front and rear pointer NONEMPTY: Only rear pointer	EMPTY: Both front and rear pointer NONEMPTY: Both front and rear pointer

Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

106	6	Consider the following implementation of linked list, which of the following lines should be inserted to complete the above code and what a function do?
		class Node
		{
		int data;
		Node next;
		}
		int function (Node t, int value)
		{
		if (t == null)
		return 0;
		if (t.data == value)
		return 1;
		return _____;
		}
			D	1	function(1), nothing	function(0), nothing	1, error	function (t.next, value), search

107	6	Which of the following statements are false?	D	1	random access of elements at linked list is not possible	arrays have better cache locality than linked list	size of linked list is dynamic and can be changed as needed	random access of elements at array is not possible

108	6	What will be the value of “count” after the following code snippet terminates?
		Consider head as the reference of the satrting Node of the Linkedlist.
		void puzzel(Node head) {
		/*
		The LinkedList is defined as:
		head.val = value of the node
		head.next = address of next element from the node
		The List is 1 -> 2 -> 3 -> 4 -> 5
		*/
		int count = 0;
		while(head.next != null) {
		count += head.val;
		head = head.next; }
		System.out.println(count);
		}
			A	1	10	5	1	2

109	6	The following function reverse() is supposed to reverse a singly linked list. There is one line missing at the end of the function.
		/* Inner class contains int data, Node next*/
		/* head_ref is a reference which points to head (or start) node of linked list */
		void reverse (node head_ref){
		node prev = null;
		node current = head_ref;
		node n1;
		while (current != null) {
		n1 = current.next;
		current.next = prev;
		prev = current;
		current = n1;
		}
		/*ADD A STATEMENT HERE*/}
		What should be added in place of "/*ADD A STATEMENT HERE*/", so that the function correctly reverses a linked list.
			A	1	head_ref=prev;	head_ref=current;	head_ref=next;	head_ref=n1;

Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

110	6	The following function reverse() is supposed to reverse a singly linked list. There is one line missing at the end of the function.
		/* Inner class contains int data, Node next*/
		/* head_ref is a reference which points to head (or start) node of linked list */
		void reverse (node head_ref){
		node prev = null;
		node current = head_ref;
		node n1;
		while (current != null) {
		n1 = current.next;
		/*ADD A STATEMENT HERE*/
		/*ADD A STATEMENT HERE*/
		current = n1; }
		head_ref =prev;}
		What should be added in place of "/*ADD A STATEMENT HERE*/", so that the function correctly reverses a linked list.
			C	1	current.next = prev; prev = current.next;	current = prev; prev = current.next;	current.next = prev; prev = current;	current = prev.next; prev = current;

111	6	In implementation of Queue data structure using Linked list, Dequeue operation of queue is equivalent to _____ operation of Linked list.	B	1	Deletion not possible	Delete at first	Deletion in descending order	Cannot be implemented using LL

112	6	void count()
		{
		Node Temp=First;
		int c=1;
		if(Temp==null)
		{
		System.out.println("Singly Linked List is Empty");
		}
		while(Temp!=null)
		{
		Temp=Temp.next; c++
		}
		System.out.println("Total Count of Nodes:"+c);
		}
		
		What will be the above count method do for a singly linked list having 12 elements.
			D	1	Print Total Count of Nodes:12	Print Total Count of Nodes:11	Print Total Count of Nodes:2	Print Total Count of Nodes:13

113	6	Consider the following function foo() which takes the first of two singly linked list as an arguments.
		
		Node foo(Node first1, Node first2)
		{
		Node final, temp;
		if(first1==null) return first2;
		if(first2==null) return first1;
		temp = foo (first1.next, first2.next);
		final = first1;
		first1.next=first2;
		first2.next=temp;
		return final;
		}
		what will be the final linked list return by foo() if executed upon following linked list?
			A	1	1,2,3,4,5,8,7,10,9,10	1,2,3,4,5,7,8,9,10	1,2,3,4,5,6,7,8,9,10	None of these

Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

114	6	Consider the following function foo() which takes the first of two singly linked list as an arguments.
		Node foo(Node first1, Node first2){
		Node final, temp;
		if(first1==null) return first2;
		if(first2==null) return first1;
		temp = foo (first1.next, first2.next);
		final = first1;
		first1.next=first2;
		first2.next=temp.data;
		return final;}
		what will be the final linked list return by foo() if executed upon following linked list?
		Linked List 1: 1→3→5→7
		Linked List 2: 2→4→6→8→9→10
			A	1	1,2,3,4,5,6,7,8,9,10	1,2,3,4,5,8,7,10,9,10	2,1,4,3,6,5,8,7,9,10	Error

115	6	Consider the following code for a singly linked list. Select the appropriate option for the missing statements. (Note – the method performs multiplication of the all the elements of a linked list.)
		void product(Node First)
		{
		Node temp=First;
		int product=1;
		while(temp!=null)
		{
		Missing Statement 1;
		Missing Statement 2;
		}
		System.out.println("Product of All elements is"+product);
		}
			C	1	product += temp.data; and temp = temp.next;	product = product + temp.data; and temp = temp.next;	product = product * temp.data; and temp = temp.next;	None

116	6	Write a JAVA program to count the number of nodes in a singly linked list		5				
117	6	Write a JAVA program to Delete the node whose value = Y		5				
118	6	Consider singly linked storage structures, Write a java program which performs an insertion at the end of a linked linear list.		5				
119	6	Write a java program for deletion in Singly Linked List.		5				
120	6	Write a function for insert in the queue implementation using Linked list.		2				
121	6	Write a method for insert at end in singly Linked list.		3				
122	6	Write a program to search an element in a linked list.		5				
123	6	Write a java program to print all odd positioned nodes from a Singly LinkedList. (Consider 1-base indexing)		3				
124	6	Write a java function named “displayfromend()” which displays the contents of the linked list from the end in singly Linkedlist		1				
125	6	Write a JAVA function that copies one linked list to another linked list. The original linked list should be starting at the pointer “head” and the newly created linked list must have starting pointer “begin”. It should work for the following test case where 3 nodes are already created having data 12, 14 and 16 in a linked list pointed by head. The newly created linked list must have the data 12 at the place where “begin” points followed by 14 and 16. Write the output of the following test case.		2				
126	6	Write a program to implement stack (PUSH, POP, DISPLAY) functions with main function using link list.		4				

Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
127	6	Write a JAVA Function for PUSH operation of Stack using Link List.		2				
128	6	Write a java Program for insertion in Ordered Singly Link list.		3				
129	6	Write a program to count the number of nodes in a singly linked list		3				
130	6	Write a Java program to determine whether all elements in a linked list are unique or not. Implement a method named insertAtLast() to insert elements in linked list, a method named areAllElementsUnique() that takes a linked list of integers as input and returns true if all elements are unique, and false otherwise.		5				